

Operating Systems – Virtualization

*Lecturer: András Millinhoffer,
milli@mit.bme.hu*

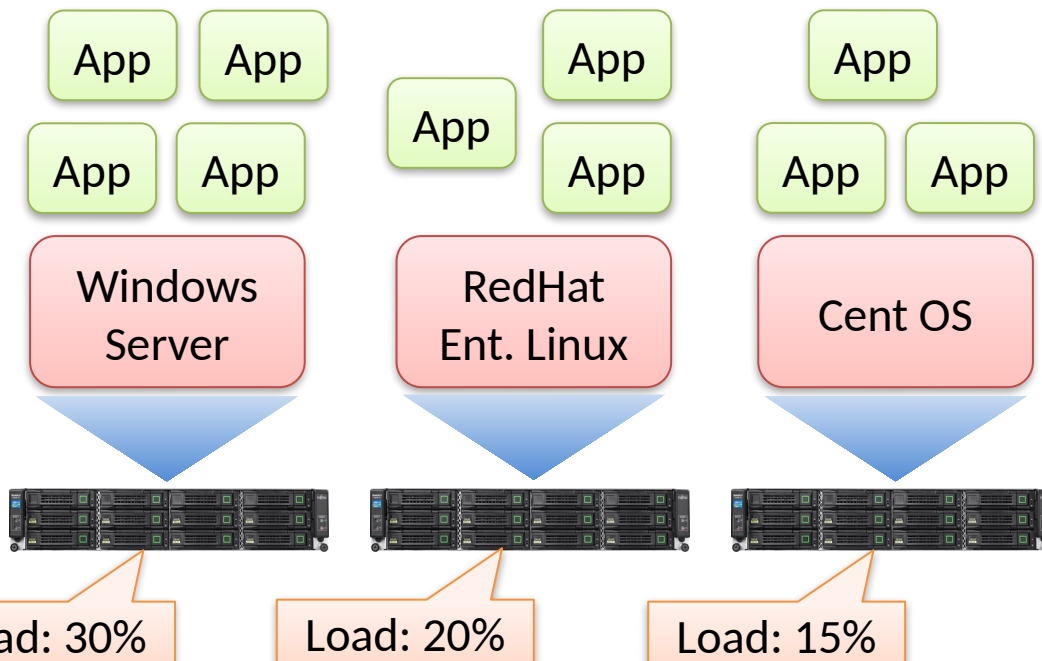
Budapest University of Technology and Economics (BME)
Department of Artificial Intelligence and Systems Engineering (MIT)

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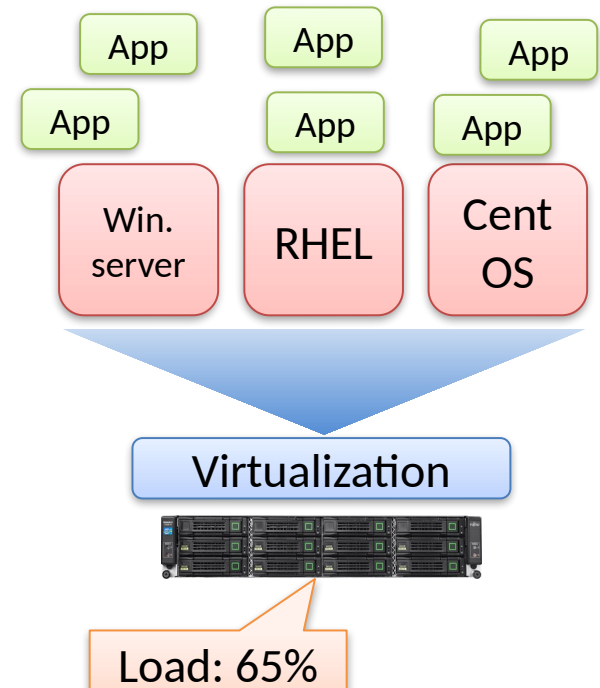
Why use virtualization?

- Separate users/tasks/OS-s
 - Depends of the type of the virtualization
- Better utilization of the HW
- Better compatibility?

No virtualization (traditional)



Virtualization

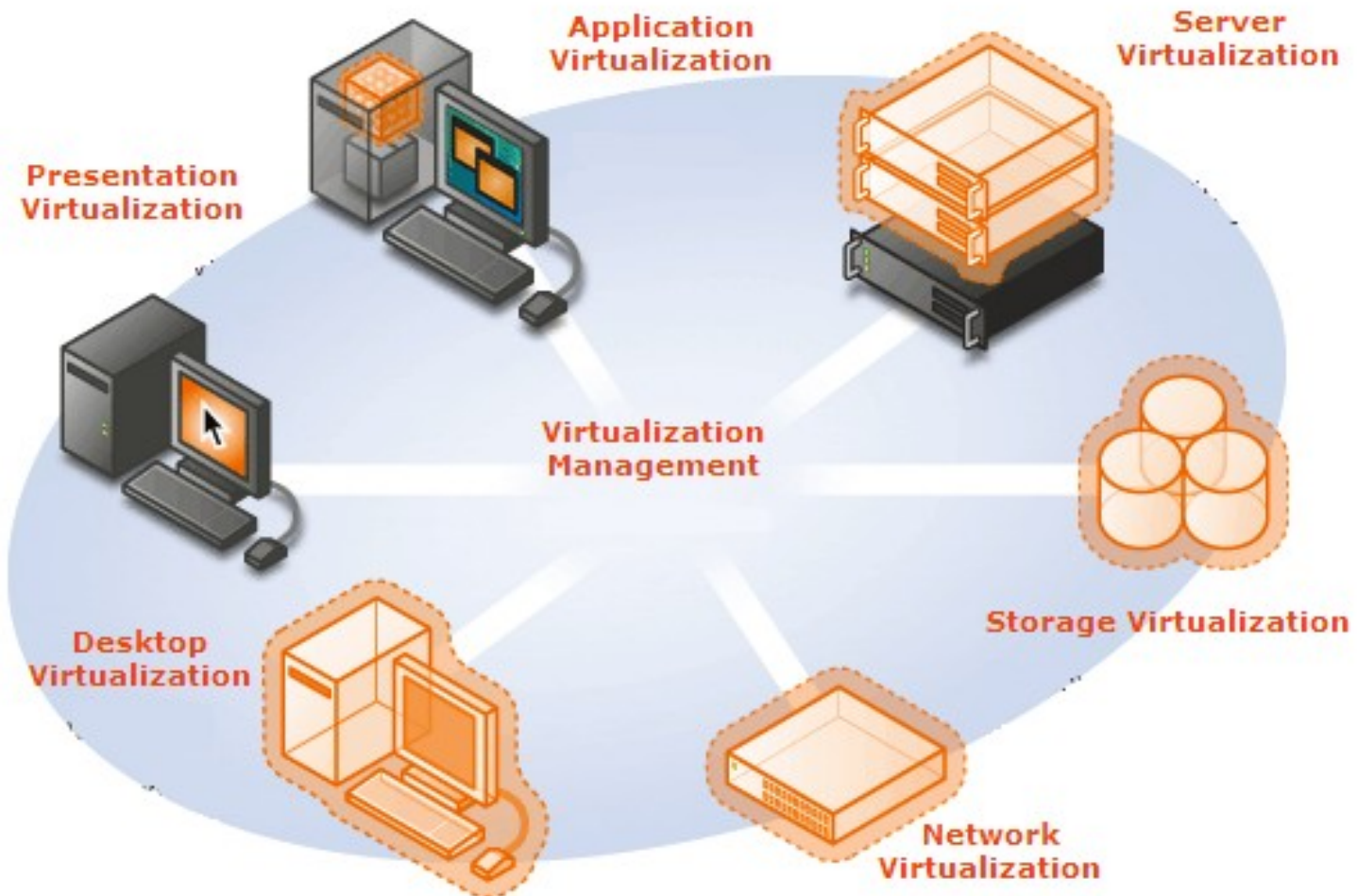


Types of virtualization

- Abstract virtual machine
 - virtual resources for the applications
 - Task separation
- Classic virtual machines
 - The HW components are shared between multiple OS-s, managed by the VMM (Virtual Machine Manager)
- Other (newer) concepts
 - OS level virtualization
 - Many users on the same OS, but they don't have to know about each other
 - Separate file systems, system libraries
 - Same kernel
 - E.g. Linux Containers
 - Application virtualization
 - Separate registry and file system for an application
 - More portable applications
 - Presentation virtualization
 - Remote monitor and input devices
 - Remote Desktop (RDP), VNC

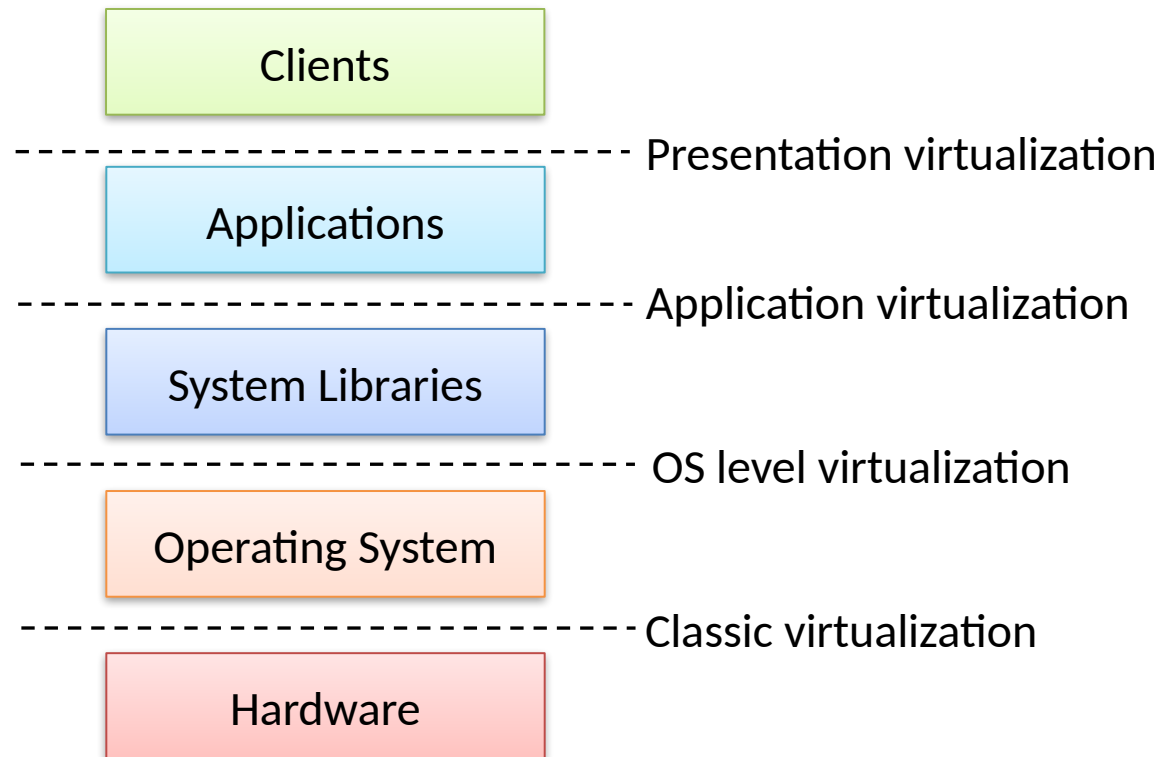
The types of virtualization in other words

- Different vendors use different terminology...

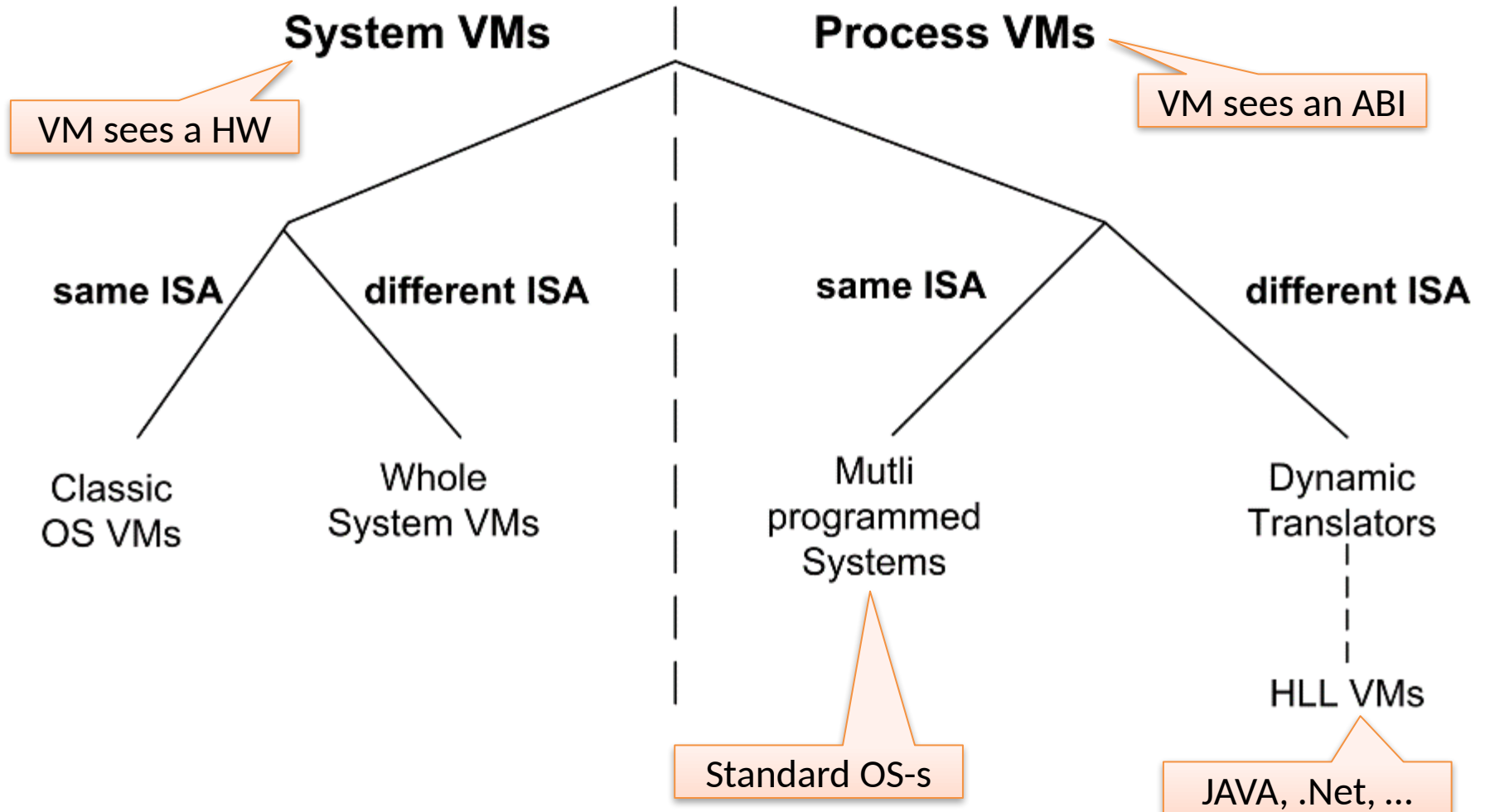


Types (levels) of virtualization

- Where we draw the line of separation?



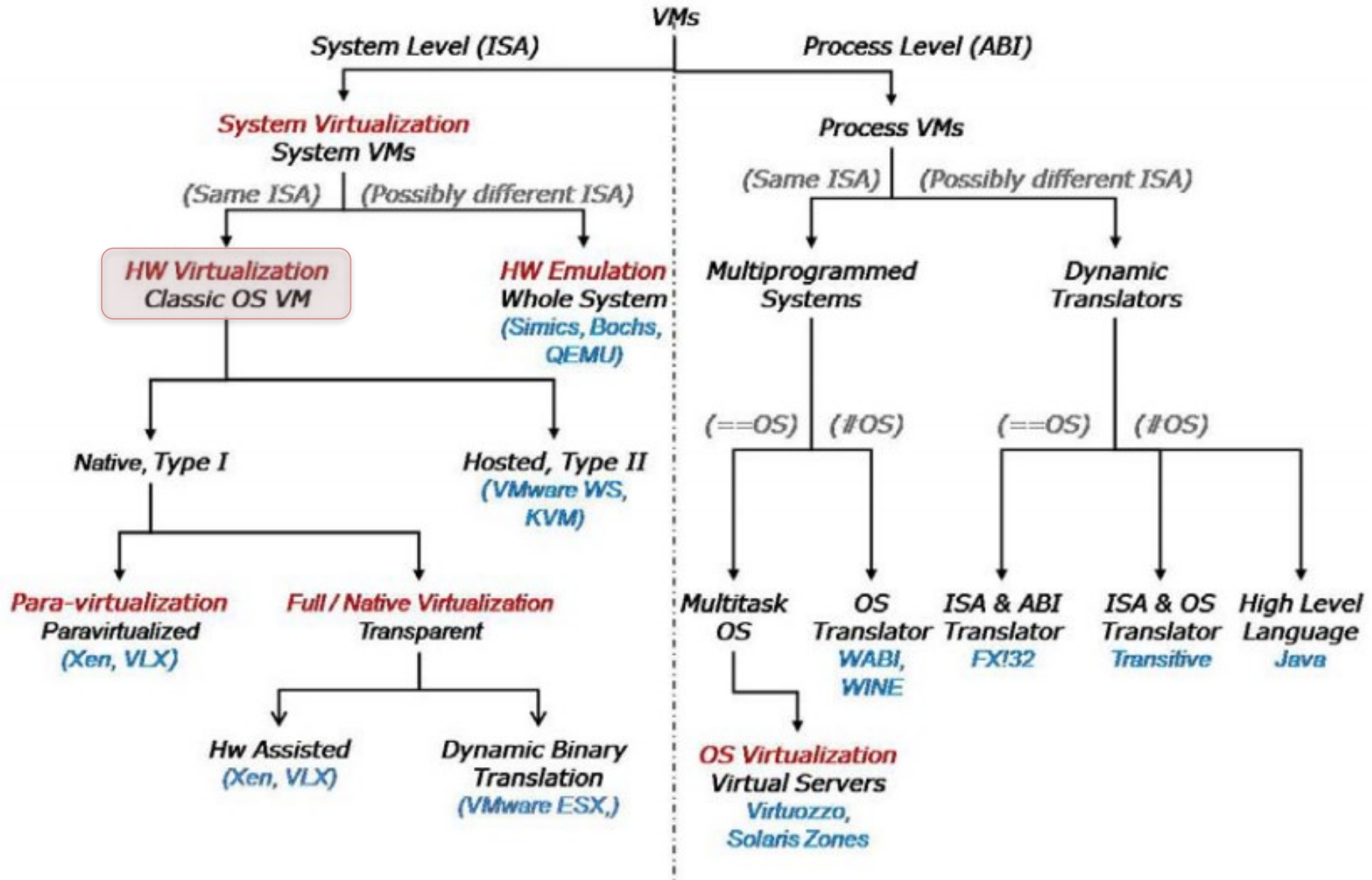
Virtual machine taxonomy*



Source: J. Smith and Ravi Nair, "The architecture of virtual machines," *IEEE Computer*, vol. 38, 2005, pp. 32-38.

* taxonomy ~ structure for presenting relationships between concepts

Virtual machine taxonomy detailed

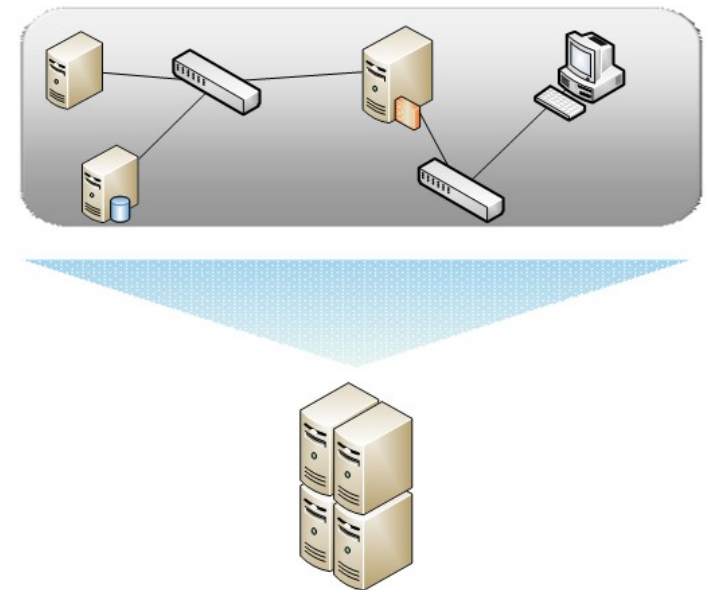


Suggested terminology

- **Platform** virtualization: virtualizing a full computer, running multiple OS on one hardware
 - Also known as: server, computer, hardware virtualization...
- Definitions:
 - **Host machine**: physical computer
 - **Guest machine**: virtual computer
 - **Virtual Machine Monitor (VMM)**: program managing the virtual machines

Why is platform virtualization good?

- Building test systems
 - Experimenting with other OS-s
 - Using a SW which is only runnable of a specific OS
- HW consolidation
- Legacy systems
 - Keeping them alive
- On-demand architectures
- High availability, disaster recovery
- Portable applications
- ...



History of platform virtualization

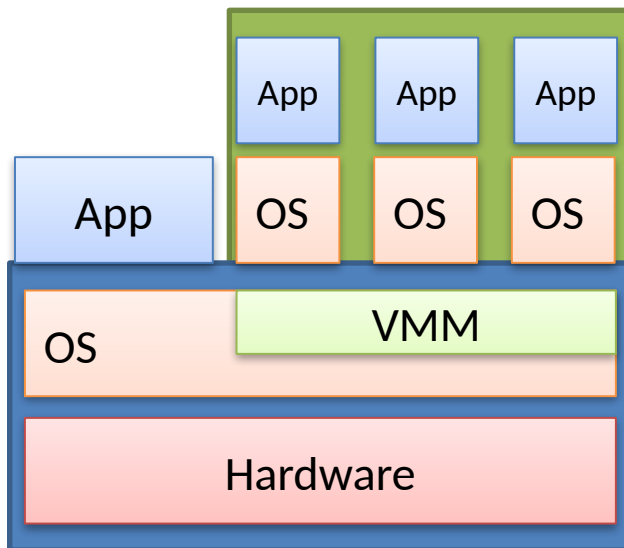
- ~1960 - IBM CP-40 system
 - in the mainframe products
- x86 virtualization
 - Seemed impossible
 - The instruction set wasn't prepared for virtualization
 - Only SW methods are possible $\Rightarrow$ can be extremely slow
 - 1997: Stanford, Disco projects
 - 1998: VMware solution
 - 2000- Other solutions
- Now:
 - HW support
 - has its own business
 - becomes widely used
 - On the enterprise level, this is the common practice



Platform virtualization

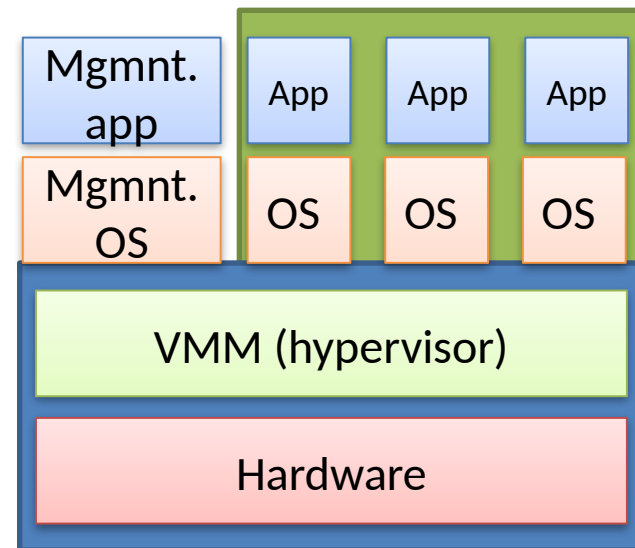
- Two main approaches

Hosted (VMM has kernel level parts)



Mainly desktop products:
VMware Workstation, Server,
Player, Oracle VirtualBox,
MS VirtualPC, KVM, UML

Bare-metal (whole VMM runs at kernel level)



Mainly server products:
VMware ESX Server, Xen
Enterprise, MS Hyper-V

GUEST

HOST

Requirements and challenges

- Requirements for a virtualization solution:
 - **Equivalence**: programs in a VM should perform indistinguishable from running on the hardware
 - **Resource control**: the VMM should handle all the physical resources
 - **Efficiency**: most of the VM's instructions should run directly on the hardware
- Challenges
 - The system have to be protected from the guest(s)
 - Not every operation is allowed
 - E.g.: HLT (Halt) instruction
 - Solution: the instructions must be monitored by the VMM
 - Privileged instructions should be handled differently - no direct execution

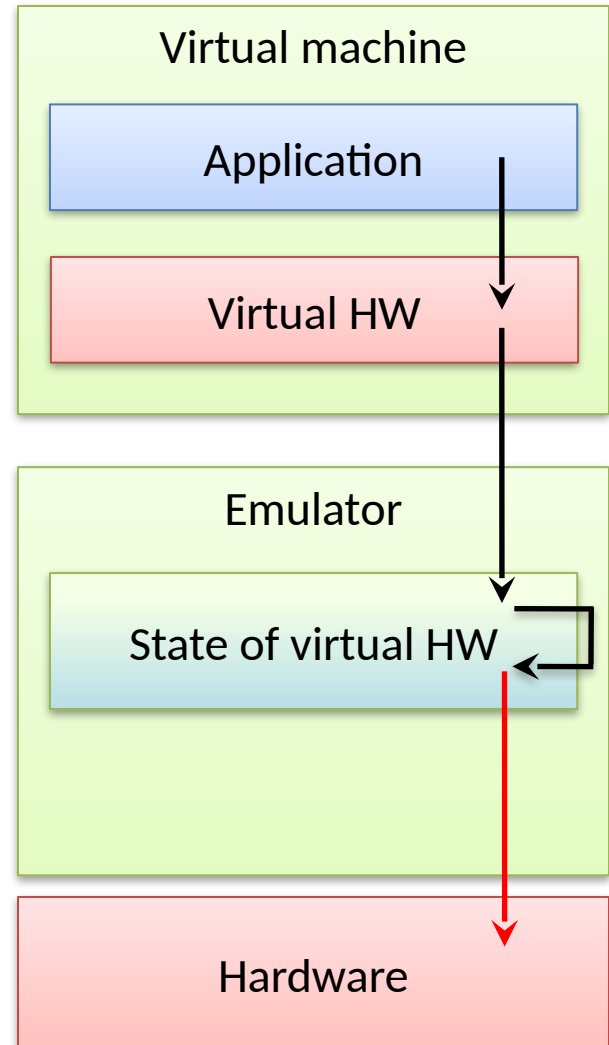
Gerald J. Popek, Robert P. Goldberg: Formal Requirements for Virtualizable Third Generation Architectures. Commun. ACM 17(7): 412-421 (1974)

Theory behind platform virtualization

- CPU virtualization
 - How to translate the instructions?
 - Every instruction is translated – emulation
 - Some instructions are translated, some executed directly
 - HW support?
 - Instruction privileges
- Memory virtualization
 - We have only 1 MMU
 - Context change between virtual machines has a high overhead
 - How to handle page tables and the TLB?
- I/O virtualization
 - How to manage a HW device? (e.g. a network adapter)
 - Use generic drivers
 - Use special virtual device drivers
 - Use special HW devices, which supports virtualization

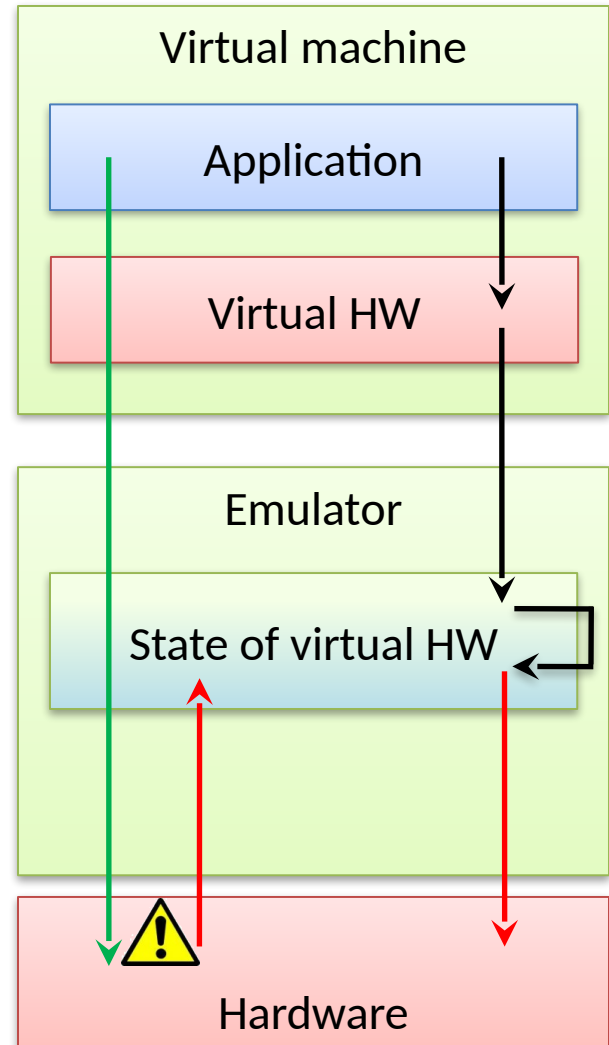
CPU virtualization – Full emulation

- The emulator
 - Stores the full state of the HW
 - Every instruction is inspected and translated, then executed
- Pro
 - Different CPU-s can be emulated
- Con
 - slow



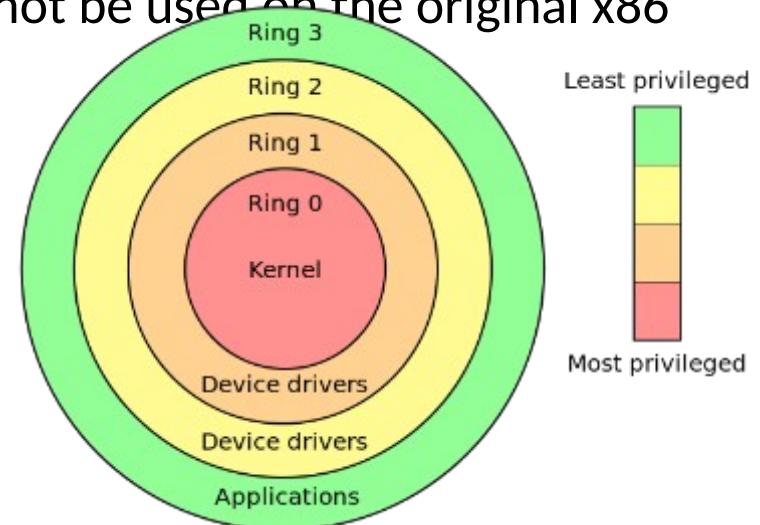
CPU virtualization – Trap and emulate

- Trap
 - HW exception handling, which resumes execution after the handler (VMM)
- HW support is required
 - Protection modes (x86 rings)
 - VM runs in a lower modes
 - Privileged instructions should cause a trap when called from a non-privileged mode



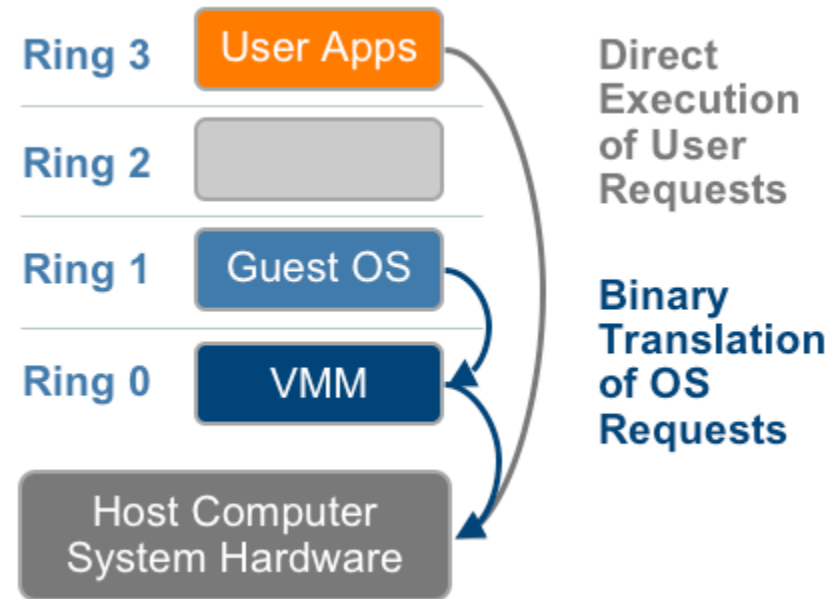
Issues with x86 virtualization

- Some architectures can be easily virtualized
 - **x86 cannot**
- From ~250 instructions 17 violate the classical requirements, e.g.
 - POPF instruction: modifies EFLAGS register
 - But if not executed in ring 0, doesn't throw an exception
- Privileged state can be detected
 - OS can detect whether it's running in a VM $\Rightarrow$ violating the equivalence requirement
- **Conclusion:** the trap & emulate method cannot be used on the original x86



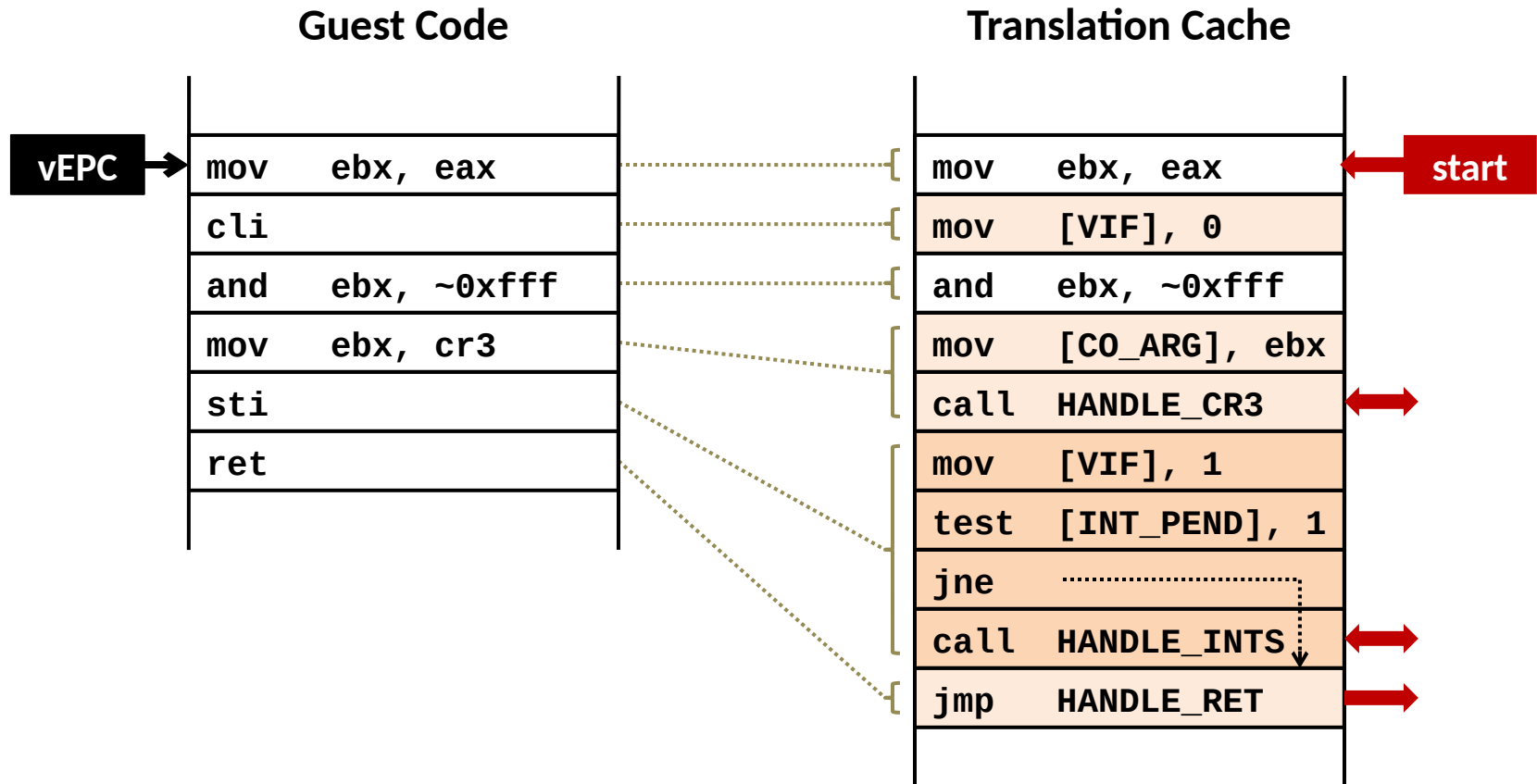
Solutions for x86 – binary translation

- Most of the instructions run directly
- The instructions are inspected in blocks
- Privileged instructions translated runtime
- Doesn't need source code
- Caches translated code
- Guest OS not aware of virtualization



Source: VMware, Understanding Full Virtualization, Paravirtualization, and Hardware Assisted Virtualization
http://www.vmware.com/files/pdf/VMware_paravirtualization.pdf

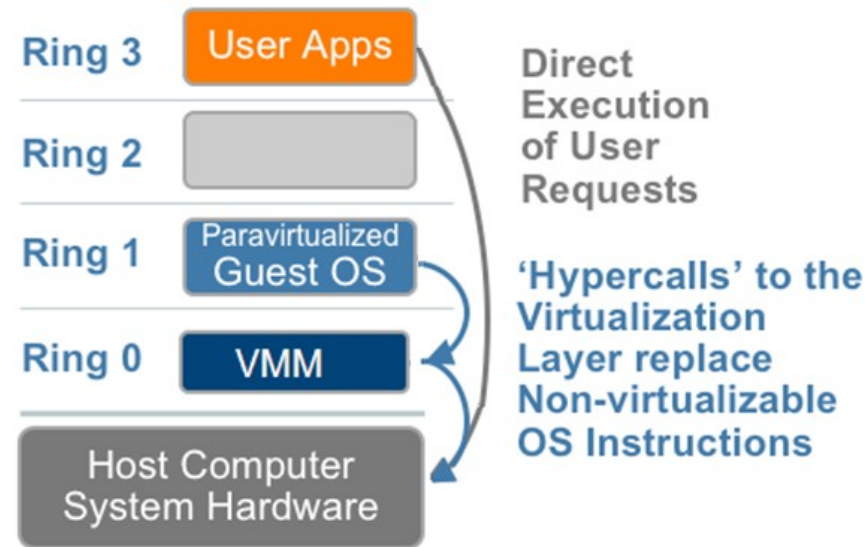
Binary translation – example



Source: Carl Waldspurger, Introduction to Virtual Machines

Solutions for x86 – paravirtualization

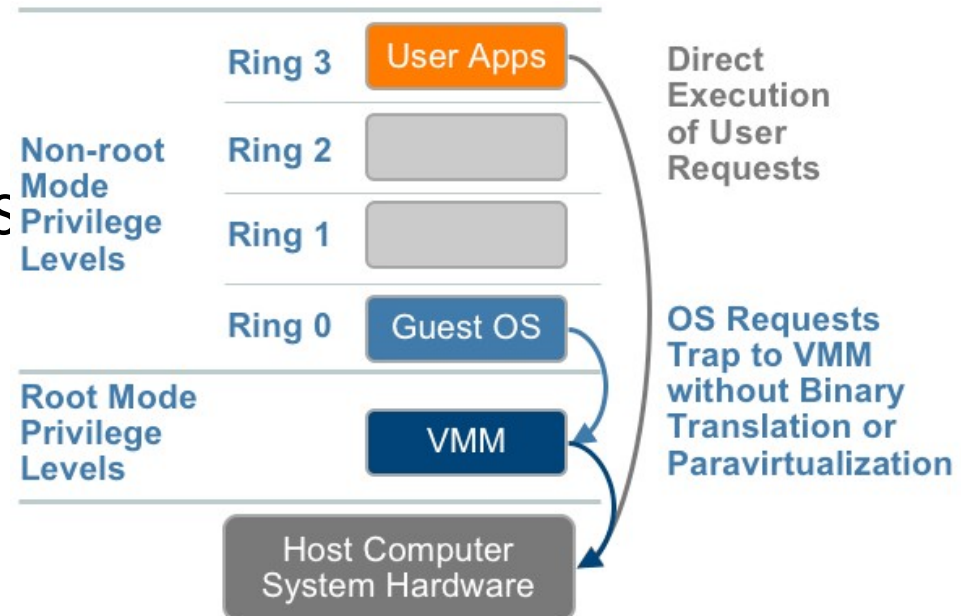
- Modifying the source of the guest OS
- Replacing “problematic” instructions
- Hypercall: calling the VMM directly



Source: VMware, Understanding Full Virtualization, Paravirtualization, and Hardware Assisted Virtualization
http://www.vmware.com/files/pdf/VMware_paravirtualization.pdf

Solutions after x86 – hardware-assisted virtualization

- ~2005: Intel Virtualization Technology (VT-x) and AMD AMD-V
- HW support: root mode, VMCS
 - Instructions: VMCALL, VMLAUNCH
- Trap & emulate now works
- Backward compatibility with the x86 ring system



Source: VMware, Understanding Full Virtualization, Paravirtualization, and Hardware Assisted Virtualization
http://www.vmware.com/files/pdf/VMware_paravirtualization.pdf

Comparison between CPU virtualization methods

- Which one is the best?
 - The answer changing constantly
 - Depends on the environment, workload
 - Most products mix several techniques
- Examples
 - 2006. VMware: [BT is better than HW assisted virtualization](#)
 - 2008. [VMware: Paravirtualization + BT is better than pure BT](#)
 - 2009.
[Comparing Hardware Virtualization Performance Utilizing VMmark v1.1](#)

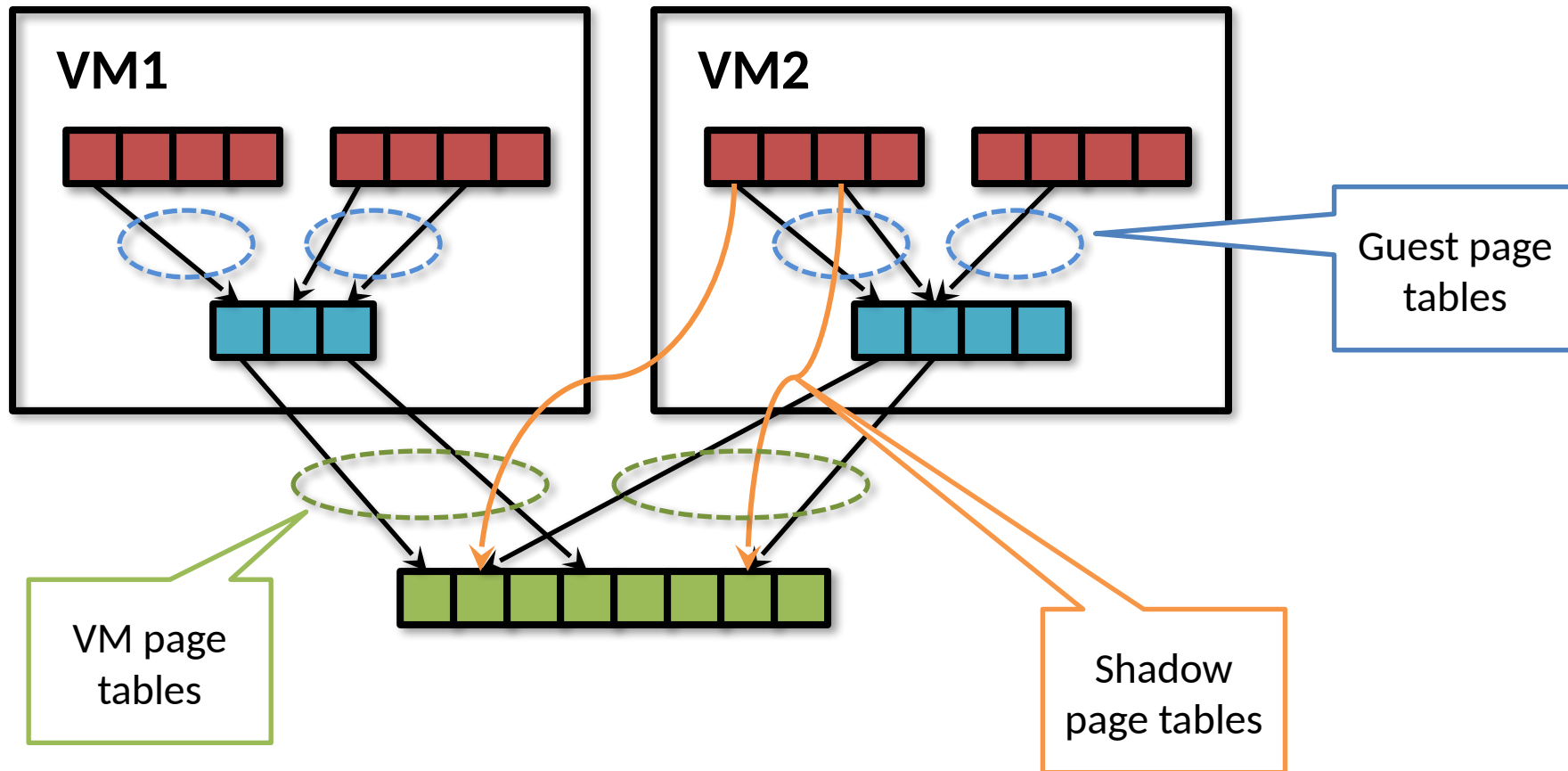
Memory virtualization (software)

- Double address translation has a high overhead

Guest virtual memory

Guest „physical” memory

Machine physical memory

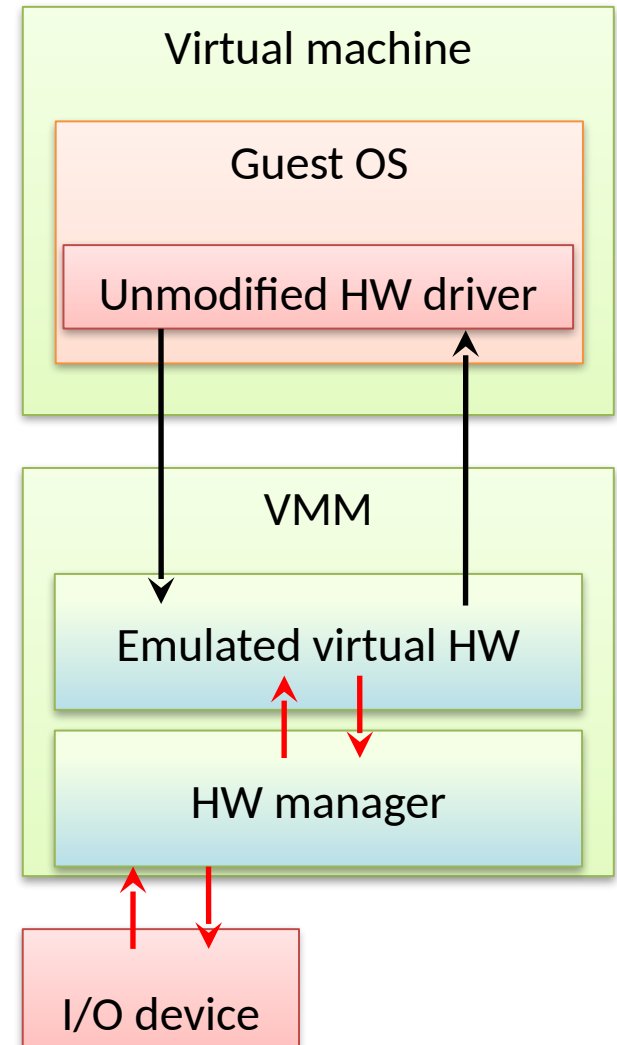


Memory virtualization – paravirtualization and hardware support

- Paravirtualization
 - Also uses shadow page tables
 - Modifying the guest OS source code
 - When the OS modifies its page tables, it should notify the VMM also
- HW support for virtualization
 - HW support in the recent CPUs
 - AMD Rapid Virtualization Indexing , Intel Extended Page Tables
 - Nested page table
 - Storing guest physical -> machines physical translation
 - Traversed by HW address translation
 - Tagging TLB entries
 - Great performance increase:
 - 2008. 04., KVM: [MMU paravirtualization is dead](#)
 - 2009., VMware: [Performance Evaluation of AMD RVI Hardware Assist](#), 42% improvement in some cases

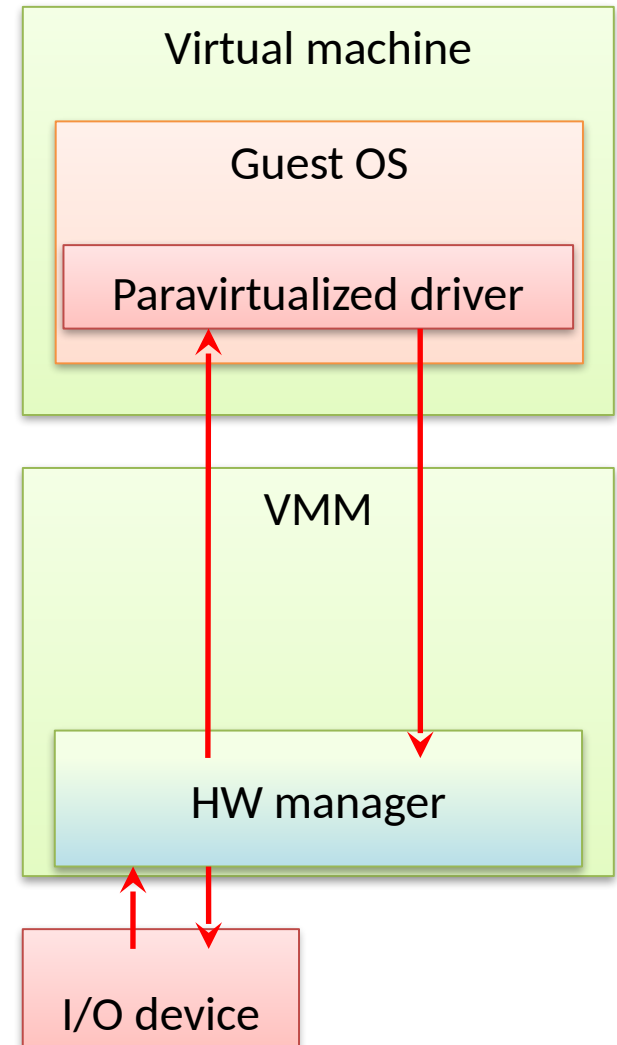
I/O virtualization – software

- Emulating the whole real communication
 - No special drivers (compatibility)
 - Can be really slow
- E.g.: many VMM-s emulate a TRIO VGA card (or other legacy type), because every OS has drivers for it



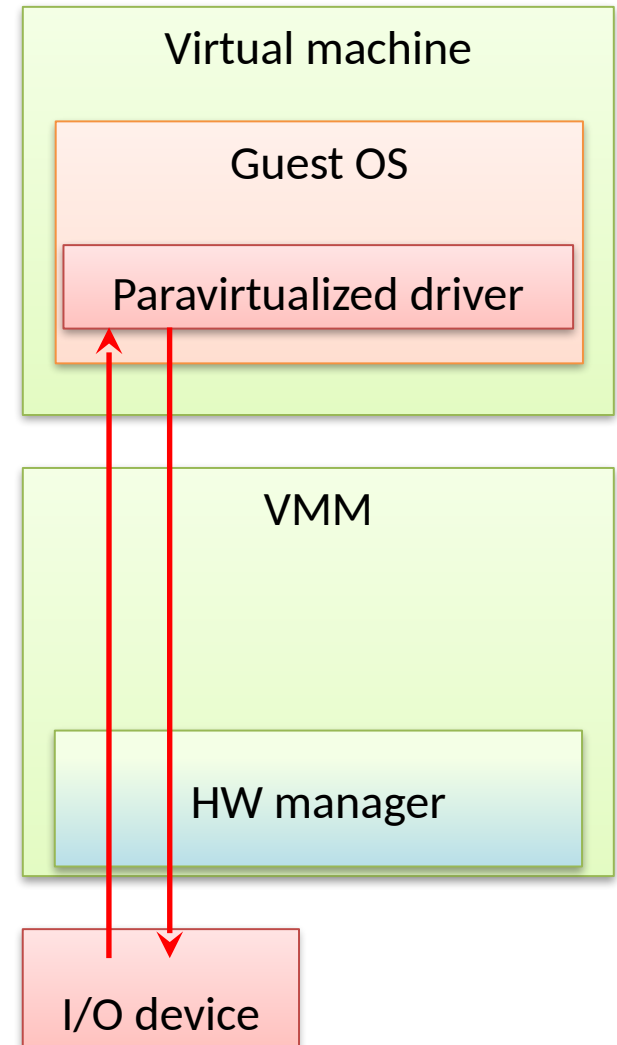
I/O virtualization – paravirtualization

- Operation
 - A special (virtual)driver is installed in the guest OS
 - Simplified calls
 - Communication through shared memory
 - Efficient operation
- Special package installed in the VM:
 - VMware Tools, Virtual PC Additions
 - **Always install these!**



I/O virtualization – hardware support

- Allowing direct access to an I/O device are not safe without a supervisor
 - Shared address range
 - More guest machines can cause conflicts
- Solutions
 - Using a HW level (fast) supervisor
 - Intel VT-d, AMD IOMMU
 - Using special I/O devices which are aware of the virtualized usage
 - PCI standard extensions: I/O Virtualization (IOV)
- Some I/O devices
 - can be shared between VMs
 - can be directly assigned to one VM
 - E.g.: GPU
 - Problems: VM context change



Products and companies

vmware®

ESXi, vSphere...



open source hypervisor

CITRIX®

XenServer, XenApp

Microsoft®

Virtual PC, Hyper-V, System Center



Solaris Containers, Oracle VM,
VirtualBox



Kernel based Virtual Machine (KVM)



mainframe, powerVM

...

Cloud computing



Types of cloud computing

IaaS

- Infrastructure-as-a-Service
- Getting a VM
- Amazon EC2, RackSpace...

PaaS

- Platform-as-a-Service
- Getting a runtime environment
 - Java container, .NET, database...
- MS Azure, Google AppEngine...

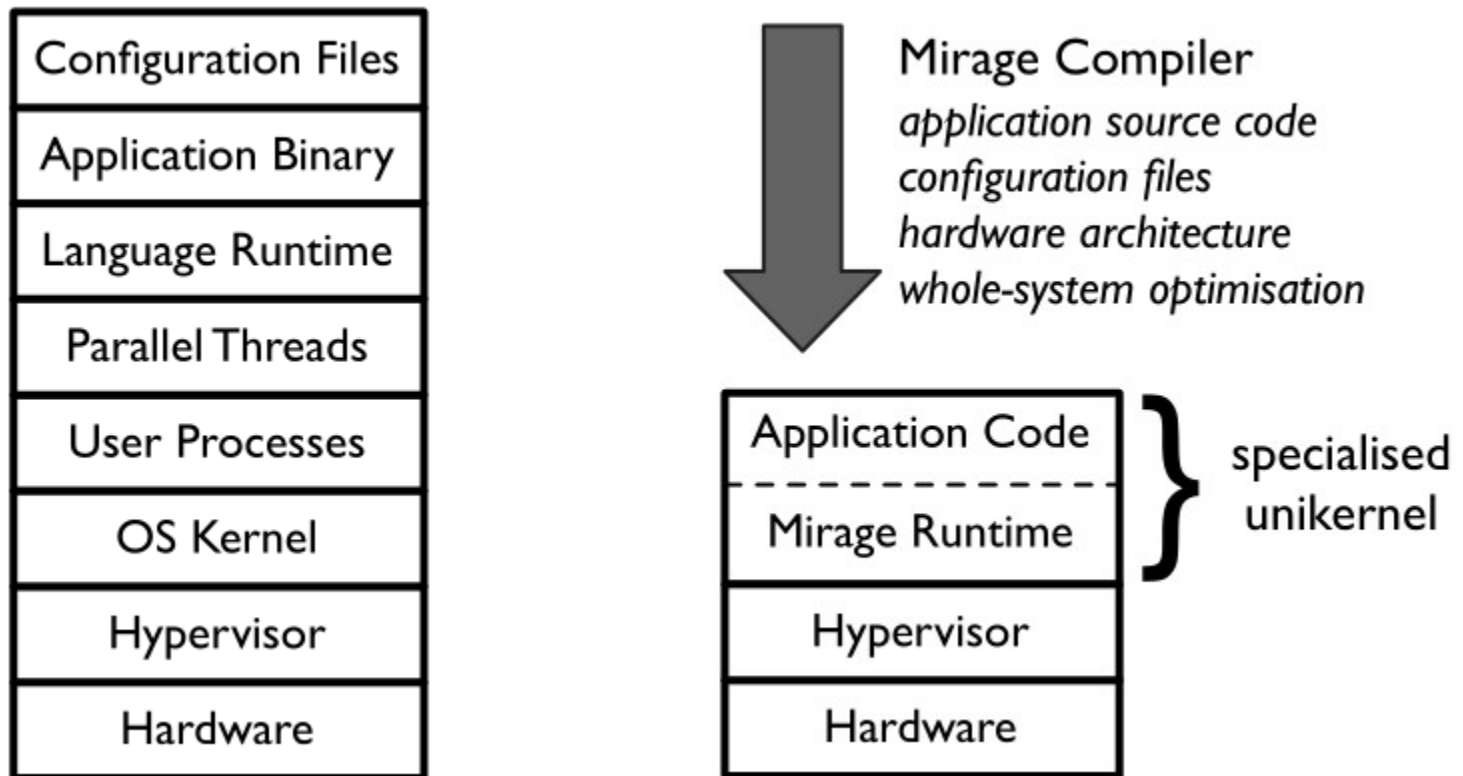
SaaS

- Software-as-a-Service
- Getting a service
- Google Docs, Microsoft Office 365, Salesforce CRM...

Current developments, future trends?

• Mirage OS

- Idea: compile a lightweight OS for a specific task
 - E.g.: for running a webserver



Summary

- Virtualization benefits
 - Better utilization, portability, sandboxing, ...
- Virtualization types (levels)
 - Platform (classic), OS, Application, Presentation
- Virtualizing
 - CPU
 - Memory
 - I/O devices
- Paravirtualization
 - Modify the existing OS or drivers to achieve greater efficiency than pure VMM SW solutions
- Cloud computing
 - IaaS, PaaS, SaaS